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FAMOS

A Multi-State Articulation Transformer Reconstructs Joint Parameters from Sparse Unordered Point Clouds Without Category Priors

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Articulated object reconstruction no longer needs a single perfect view or a category-specific prior. *FAMOS* (Feed-forward Articulation Modeling from Sparse Observations) accepts an unordered set of partial point clouds of the same object in different joint states, aggregates articulation evidence across views with a Multi-state Articulation Transformer, and predicts movable-part segmentation plus full joint parameters in a single forward pass—outperforming the best prior feed-forward method by up to 27.8 points on three standard benchmarks.

AT A GLANCE

Problem: Existing feed-forward methods infer articulation from a single observation and rely on strong category priors that fail on novel instances and unusual joint configurations.

Why it matters: Robots that can estimate joint parameters of arbitrary objects from a few casual scans would generalise to unstructured environments without per-object calibration.

Method: Multi-state Articulation Transformer with alternating state-wise and global attention; Observed Articulation Span (OAS) objective to exploit motion diversity; procedural synthetic data generation to overcome dataset scarcity.

Result: +8.1, +27.8, +21.5 over Particulate on PartNet-Mobility, ACD, and ArtiCraft-10K in joint axis error.

Evidence: Evaluation on three established articulated-object benchmarks with ablations over attention variants, OAS, and observation count.

The Big Picture

Articulated objects—drawers, cabinets, scissors, robots—have rigid parts connected by kinematic joints. Knowing the joint type, axis, origin, and motion range is prerequisite for any downstream manipulation planner. The classical approach requires dense RGBD scans or repeated observations across many configurations; modern feed-

forward methods aim to infer joint parameters in a single forward pass but have so far been limited to one observation and one category at a time.

FAMOS abandons both restrictions. Its input is a sparse, unordered set of partial point clouds $\{X_n\}_{n=1}^N$ capturing the object in N different joint states; its output is per-point part membership and per-part joint parameters. Because the model sees the object in multiple configurations, it can directly observe how parts move relative to one another, making the problem substantially easier without requiring explicit motion correspondence.

Why Existing Approaches Fall Short

Optimization-based methods (A-SDF, DITTO) fit a parametric model to each observation at test time. They are slow (minutes per object), sensitive to initialisation, and require category-specific shape templates—none of which generalise to novel object categories.

Single-view feed-forward methods (Particulate, ArtSplat) amortise this cost by learning a category-level prior during training. Their critical weakness is that a single partial observation reveals only part of the object in one configuration. Without multi-view evidence, the model must hallucinate the motion range from prior experience, producing large errors on out-of-distribution instances where the prior is wrong.

Neither exploits the structure of multi-state observations. When $N > 1$ partial scans are available, each pair of scans constrains the joint axis: the rigid-body transformation between a moving part’s point clouds in two states lies in the joint’s motion manifold. FAMOS is the first feed-forward model designed to reason jointly over all observations.

Core Mechanism

Point encoding. Each point cloud X_n is encoded independently by a shared PointNet-style encoder ϕ into a feature map $F_n \in \mathbb{R}^{T \times D}$ ($D = 768$). $K = 16$ learnable part query vectors $\mathbf{q}_k$ are initialised for each forward pass.

Multi-state Articulation Transformer (MAT). The $L = 6$ transformer layers each alternate between two attention modes:

State-wise attention: For observation n , part queries cross-attend to point features of that observation only: $\mathbf{q}_k^{(n)} \leftarrow \text{Attn}(\mathbf{q}_k, F_n)$.

Global attention: All part queries from all observations attend jointly: $\{\mathbf{q}_k^{(n)}\}_{n,k} \leftarrow \text{Attn}(\{\mathbf{q}_k^{(n)}\}_{n,k}, \{\mathbf{q}_k^{(n)}\}_{n,k})$.

Global attention is the key innovation: it lets the model compare articulation cues across observations and identify which joints are active in which views.

Decoding. Separate MLP heads read each final query $\mathbf{q}_k$ and output joint axis, origin, range, and part membership logits for every input point.

Technical Formulation

The Observed Articulation Span (OAS) objective augments standard joint- parameter regression. Let $\theta_n^{(m)}$ be the joint angle of part m in observation n . The observed motion range is:

$$\Delta_m = \max_n \theta_n^{(m)} - \min_n \theta_n^{(m)}.$$

A regression head predicts $\hat{\Delta}_m$ from the part query $\mathbf{q}_k$:

$$\mathcal{L}_{\text{OAS}} = \sum_m \left(\hat{\Delta}_m - \Delta_m \right)^2.$$

The full loss combines part segmentation (Hungarian-matched cross-entropy), joint axis regression (cosine loss), origin regression (ℓ_2), motion range regression (ℓ_2), and OAS:

$$\mathcal{L} = \lambda_{\text{seg}} \mathcal{L}_{\text{seg}} + \lambda_{\text{axis}} \mathcal{L}_{\text{axis}} + \lambda_{\text{origin}} \mathcal{L}_{\text{origin}} + \lambda_{\text{range}} \mathcal{L}_{\text{range}} + \lambda_{\text{OAS}} \mathcal{L}_{\text{OAS}}.$$

Joint axis loss uses the unsigned cosine:

$$\mathcal{L}_{\text{axis}} = 1 - |\cos(\hat{\mathbf{a}}_m, \mathbf{a}_m)|.$$

Learning or Inference Procedure

1. Encode each observation independently with the shared PointNet encoder to obtain $\{F_n\}$.
2. Initialise K learnable part queries $\{\mathbf{q}_k\}$.
3. Run L MAT layers alternating state-wise and global attention.
4. Decode part queries via MLP heads into joint parameters and per-point part membership logits.
5. OAS head is used only during training; at inference, joint parameters are read from the main decoders.

Training data mixes real PartNet-Mobility instances with procedurally generated synthetic assets. The procedural generator samples random kinematic trees (depth 1–3), assigns primitive or CAD-derived meshes, and renders partial point clouds from multiple random configurations, yielding unlimited training variety.

What the Guarantee Says

FAMOS provides no formal approximation bound, but the paper demonstrates empirically that axis error drops monotonically as N increases from 1 to 4 observations on every benchmark. The largest single gain is from $N=1$ to $N=2$ (4.3 geodesic degrees), confirming that even one additional view substantially constrains the joint parameter space.

Experimental Findings

Three benchmarks cover in-distribution and out-of-distribution object types:

- **PartNet-Mobility:** 45 categories, 2,346 objects.
- **ACD:** Articulated CAD objects with novel geometries.
- **ArtiCraft-10K:** Large community-contributed dataset.

With 4 input observations (geodesic axis error in degrees, lower is better):

Method	PN-Mob	ACD	ArtiCraft
A-SDF	24.1	31.4	38.7
DITTO	21.3	28.6	35.2
Particulate	18.4	22.5	29.1
FAMOS	10.3	16.9	7.6

FAMOS’s biggest gain is on ArtiCraft (21.5 point improvement), which contains the widest variety of object categories and thus most penalises methods that depend on category-specific shape priors.

Ablations and Interpretation

Global attention is essential: Removing global attention and using only state-wise attention degrades axis error by 3.6 geodesic degrees on PartNet-Mobility; without cross-view comparison the model cannot distinguish joint axes from part-shape variation.

OAS objective matters: Removing OAS increases axis error by 2.9 degrees; the model conflates partial observation with small range, mis- estimating joint limits.

Synthetic data is the enabler: Removing procedural synthesis while keeping real data causes a 4.2-degree degradation; the real datasets alone are too small and category-limited for the multi-view model to generalise.

$N = 1$ is a competitive single-view baseline: Even with $N = 1$, FAMOS outperforms Particulate by 1.7 degrees on PartNet-Mobility, because the MAT architecture is a better single-view reasoner than prior point-cloud networks.

Reference

Kevin Qu, Stefan Ainsworth, Jiahao Wang, Gordon Wetstein, Qianqian Wang. **FAMOS: Feed-Forward 3D Articulation Modeling from Sparse Observations.** arXiv:2609.20817, September 2026. <https://arxiv.org/abs/2609.20817>

Keywords: agentic reinforcement learning • on-policy distillation • GRPO • adaptive training schedule

RetireOPD

Adaptive Retirement of Teacher Distillation Unlocks an 18-Point Gain for Multi-Turn Agentic RL on ALFWorld and WebShop

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Teacher distillation can harm as much as it helps when training agentic reinforcement learning—unless the student knows when to stop listening. *RetireOPD* separates a skill-conditioned teacher from a skill-free student, trains them in sequence, and lets the student autonomously retire distillation once it has absorbed what the teacher can teach. The result: up to +18.8% success rate on ALFWorld and +19.0% on WebShop over pure RL across three model scales.

AT A GLANCE

Problem: Multi-turn agents trained with sparse-reward RL receive weak learning signals; on-policy distillation (OPD) from a privileged teacher helps early but harms later when the teacher becomes a bottleneck.

Why it matters: Web and household agents must make long chains of tool calls; sparse end-of-episode rewards provide almost no signal for individual action improvements.

Method: Pre-train a skill-conditioned teacher with GRPO; jointly train a skill-free student with RL + OPD; retire OPD automatically once the student’s discrepancy stops shrinking and it reaches a target success fraction of the teacher.

Result: +14.1–18.8% on ALFWorld; +11.8–19.0% on WebShop vs. RL-only at three model scales (1.5B, 3B, 7B).

Evidence: Ablations over teacher pre-training, retirement triggers, and α threshold sensitivity; comparisons vs. SFT, GRPO, continuous OPD, and SAGE-OPD.

The Big Picture

Long-horizon agentic tasks—booking a product on a website, completing a household chore sequence—require LLMs to make dozens of sequential decisions before receiving any feedback. Reinforcement learning can discover

effective strategies from this sparse reward but converges slowly because the credit assignment problem is severe: which of the 40 actions in a successful trajectory were actually critical?

On-policy distillation (OPD) offers a shortcut: a teacher that has privileged information (a task plan, a ground-truth action sequence, oracle environment state) generates dense token-level supervision, and the student imitates the teacher while simultaneously learning from RL. Prior work treats OPD as a fixed co-training objective. Retire-OPD shows this is wrong: the benefit of distillation is not constant across training, and forcing the student to keep imitating a teacher it has already surpassed in some dimensions causes active regression.

Why Existing Approaches Fall Short

Pure RL (GRPO) trains with a clipped surrogate policy gradient but has no mechanism to bootstrap from the teacher’s privileged knowledge. The agent must discover strategies entirely from environment feedback, requiring many more episodes to converge.

Continuous joint OPD + RL combines distillation and RL throughout training. Two problems arise: (1) a teacher that was only marginally better than the student at initialisation, because privileged information alone does not make a teacher reliable in agentic settings where the teacher must also plan correctly, provides harmful early supervision; (2) later in training, the student explores behaviours beyond the teacher’s repertoire that OPD actively suppresses.

SAGE-OPD uses a predefined schedule to taper OPD linearly, which is better than constant OPD but cannot adapt to the variable rate at which different model scales or task complexities saturate distillation.

Core Mechanism

Phase 1: Teacher pre-training. A skill-conditioned teacher $\pi_T(a_t | s_t, z)$ is initialised from the base LLM and fine-tuned with GRPO on environment rewards augmented by a skill-following term that rewards use of the provided privilege z (e.g., sub-task labels, reference trajectories). This phase ensures the teacher is reliably better than a skill-free policy before distillation begins.

Phase 2: Student joint training. A skill-free student $\pi_S(a_t | s_t)$ is trained with:

$$\mathcal{L}_S = \mathcal{L}_{\text{GRPO}}(\pi_S) + \beta \cdot \mathcal{L}_{\text{OPD}}(\pi_T, \pi_S),$$

with teacher parameters frozen.

Adaptive Retirement. At each training step, two signals are monitored:

- *Discrepancy:* $\Delta_t = \text{KL}(\pi_T || \pi_S)$ averaged over recent rollouts.

- *Success gap*: $\rho_t = R_S^t/R_T$, the student’s rolling success rate relative to the teacher’s fixed rate.

Retirement fires when:

$$\Delta_t - \Delta_{t-k} < \epsilon_\Delta \quad \text{AND} \quad \rho_t \geq \alpha.$$

After retirement, $\beta=0$ and only $\mathcal{L}_{\text{GRPO}}$ is used.

Technical Formulation

The GRPO loss is:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}_\tau \sum_t \min \left(\frac{\pi_S(a_t)}{\pi_{\text{ref}}(a_t)} \hat{A}_t, \text{clip} \left(\frac{\pi_S}{\pi_{\text{ref}}}, 1 \pm \epsilon \right) \hat{A}_t \right)$$

where $\hat{A}_t$ is the group-relative advantage and π_{ref} is the reference (SFT checkpoint or retirement-point snapshot).

The OPD loss is:

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_\tau \sum_t \text{KL}(\pi_T(\cdot | s_t, z) \parallel \pi_S(\cdot | s_t)).$$

The teacher pre-training loss adds a skill-following reward:

$$r_T = r_{\text{env}} + \lambda_z \cdot \mathbf{1}[\pi_T \text{ uses } z].$$

Default hyperparameters: rolling window $k = 100$, convergence threshold $\epsilon_\Delta = 0.01$, success-fraction threshold $\alpha = 0.85$, distillation weight $\beta = 0.5$.

Learning or Inference Procedure

1. Pre-train skill-conditioned teacher with GRPO + skill reward for T_{teacher} steps until teacher success rate saturates.
2. Initialise student from base LLM (same checkpoint as teacher init).
3. Train student jointly with $\mathcal{L}_{\text{GRPO}} + \beta \mathcal{L}_{\text{OPD}}$.
4. At each step compute Δ_t and ρ_t ; fire retirement condition if met; set $\beta=0$ and continue RL-only.
5. At inference: deploy skill-free student π_S . Teacher and privilege z are not needed.

What the Guarantee Says

No formal guarantee is provided. The paper motivates retirement theoretically: under a noisy oracle model where the teacher is correct with probability $p > 0.5$ on each token, continuing OPD past the saturation point has *negative* expected alignment between the OPD gradient and the true reward gradient. Retirement is designed to detect this saturation empirically before the misalignment damages performance.

Experimental Findings

Evaluated on Qwen2.5 at three scales (1.5B, 3B, 7B):

Method	ALFWorld	WebShop
SFT	54.2	59.3
GRPO only	67.1	68.4
Continuous OPD+RL	71.3	72.1
SAGE-OPD	73.8	74.5
RetireOPD	85.9	87.4

Success rate (%), 7B scale. RetireOPD exceeds SAGE-OPD by 12.1% on ALFWorld and 12.9% on WebShop, confirming that adaptive retirement, not just scheduled tapering, is the key.

Ablations and Interpretation

No teacher pre-training: -8.3% on ALFWorld; the teacher provides unreliable supervision at initialisation, corrupting early student gradients.

No retirement (continuous OPD): -14.6% ; continued OPD past saturation actively suppresses productive student exploration.

Discrepancy-only retirement: Premature retirement at 1.5B scale (-4.1%); the student retires before reaching competence.

Success-gap-only retirement: Over-extended distillation at 7B scale (-2.7%); threshold is met late after OPD stops improving.

α sensitivity: $\alpha \in [0.80, 0.90]$ is robust; $\alpha < 0.70$ causes premature retirement; $\alpha > 0.95$ delays retirement past the benefit window.

Reference

Yan Yu, Zhengxi Lu, Yizhou Liu, Yichen Pan, Aozhe Wang, Qipeng Chen, Hua Yang, Wenqi Zhang, Weiming Lu, Qianglong Chen, Yongliang Shen. **RetireOPD: Self-Retiring On-Policy Distillation for Agentic Reinforcement Learning**. arXiv:2609.20784, September 2026. <https://arxiv.org/abs/2609.20784>

Keywords: hierarchical robot planning • vision-language models • agentic distillation • stage-transition detection

StageGuard

Three-Layer Agentic Distillation Teaches a Compact VLM to Decide When Robots Should Switch Subtasks—at Real-Time Speed

By Wenbo Cui, Jing Xu, Rui Zhao, Yunchao Liu, Chaobin Ye, Xinyu Zhang & Minghui Tan (South China University of Technology; Tencent Robotics X)

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The bottleneck in long-horizon robot planning is not the skills themselves but knowing when to stop one and start the next. *StageGuard* distills structured three-layer reasoning from a large cloud VLM into a compact local model that runs at 11 Hz, achieves F1 = 88.4 on stage-transition detection—7.2 points above the best prior method—and improves full-task success rate by 9.7 percentage points on BEHAVIOR-1K.

AT A GLANCE

Problem: Stage-transition decisions in hierarchical robot planning require human-designed completion checkers that fail under perceptual noise; large VLMs can reason about completion but are misaligned with subtask criteria and too slow for real-time control.

Why it matters: Errors in stage transitions cascade: a premature transition moves to the next skill before the current one is complete, and a missed transition stalls the robot indefinitely.

Method: A teacher VLM generates structured three-layer traces (state, task, decision) from demonstration trajectories; a compact student VLM is fine-tuned on these traces and deployed on-robot.

Result: F1 = 88.4 (sim), 87 ms latency, 67.1% task success rate on BEHAVIOR-1K; validated on a real Franka Panda.

Evidence: Comparison with rule-based checkers, zero-shot GPT-4V, and fine-tuned VLM without structured reasoning; ablations over each reasoning layer and observation history length.

The Big Picture

Hierarchical robot planners decompose complex tasks (e.g., “load the dishwasher and close it”) into a sequence of subtasks (grasp plate, place plate, grasp glass, ..., close door), each executed by a dedicated low-level skill policy. The stage-transition monitor decides when to terminate the current skill and invoke the next.

This is a harder problem than it appears. Low-level skill policies produce smooth trajectories that may overshoot, stall, or partially succeed. The robot may not be able to observe all relevant state directly. Transition criteria that are trivially stated in natural language (“the drawer is fully closed”) require fine-grained spatial reasoning under occlusion and varied lighting.

Current solutions are hand-crafted rule sets: threshold on end-effector position, contact detection, object bounding-box IoU. These rules are brittle, task-specific, and require re-engineering for each new subtask.

Why Existing Approaches Fall Short

Rule-based completion checkers must be designed per subtask by a robotics engineer, do not generalise to novel objects or environments, and fail under sensor noise. Their F1 on BEHAVIOR-1K is only 71.3.

Cloud VLMs (GPT-4V, Qwen-VL-Max) can interpret diverse scene images but have three problems for this use case: (1) their query-answer format is not calibrated to subtask-completion criteria specifically; (2) they operate at 1–3 seconds per query vs. 10 Hz control loops; (3) they require cloud access, making them unsuitable for offline or safety-critical deployments.

Fine-tuned VLMs without structured reasoning achieve F1 = 81.6 but have a flat, one-layer reasoning process: they output binary decisions without decomposing state observations, task semantics, or decision rationales separately, limiting their robustness.

Core Mechanism

The teacher VLM processes each timestep observation through three sequential prompting stages that build a structured trace ($z_{\text{state}}, z_{\text{task}}, z_{\text{decision}}, d_t$):

State layer. Given observation history $o_{t-H:t}$, the teacher extracts a structured state summary: current object poses, grasp contacts, spatial relations, and completion indicators (e.g., “cup rim 2 cm above shelf surface, no contact”).

Task layer. Given z_{state} and the current/next subtask pair (T_i, T_{i+1}) , the teacher maps state facts to subtask completion criteria: which aspects of z_{state} determine whether T_i is done, and what preconditions T_{i+1} requires.

Decision layer. Given z_{task} , the teacher issues a binary transition verdict $d_t \in \{0, 1\}$ with a natural-language rationale z_{decision} .

A compact local student VLM (3B parameters) is fine-tuned on $(o_{t-H:t}, T_i, T_{i+1}) \rightarrow (z_{\text{decision}}, d_t)$ pairs generated by running the teacher on all demonstration trajectories.

Technical Formulation

Formally, the monitor observes $o_t = (I_t, \mathbf{q}_t, \mathbf{p}_t)$ (RGB image, joint configuration, end-effector pose) with history window H . The transition decision:

$$d_t = f(o_{t-H:t}, T_i, T_{i+1}) \in \{0, 1\}.$$

The student produces a soft confidence from its last-layer hidden state:

$$c_t = \sigma(w^\top h_t), \quad h_t = \text{Student}(o_{t-H:t}, T_i, T_{i+1}),$$

triggering a transition when $c_t \geq \tau$ (set by validation).

The student training loss combines binary cross-entropy on the decision and a token-level alignment loss on the compressed explanation:

$$\mathcal{L}_S = \mathcal{L}_{\text{CE}}(d_t, \hat{d}_t) + \lambda \mathcal{L}_{\text{align}}(z_{\text{decision}}, \hat{z}_{\text{decision}}).$$

The alignment loss is a token-level cross-entropy weighted by teacher confidence:

$$\mathcal{L}_{\text{align}} = - \sum_j \hat{p}_j \log p_j^{\text{student}},$$

where $\hat{p}_j$ are teacher token probabilities for explanation token j .

Learning or Inference Procedure

1. Collect human demonstration trajectories for target task set.
2. Run cloud teacher VLM (offline) to produce structured traces for all trajectory timesteps.
3. Fine-tune compact student VLM on trace-decision pairs.
4. Deploy student on-robot: at each 10 Hz control step, pass recent observation and subtask context; trigger transition if $c_t \geq \tau$.
5. Log student explanation text for safety records and debugging.

Training uses $\lambda = 0.3$, history $H = 5$, threshold $\tau = 0.55$. Teacher traces are generated once and cached; student fine-tuning takes approximately 4 hours on a single A100.

What the Guarantee Says

StageGuard provides no formal guarantees. Empirically, the student’s F1 is within 2.3 points of the teacher’s F1 on held-out BEHAVIOR-1K trajectories, while inference latency drops from 1,820 ms (cloud VLM) to 87 ms (local student), enabling real-time monitoring at 11 Hz. Real-robot experiments on five Franka Panda tasks confirm the improvements transfer beyond simulation.

Experimental Findings

Method	F1	Lat. (ms)	Task SR
Rule-based	71.3	2	48.2
GPT-4V zero-shot	79.1	1820	52.7
VLM FT (no reasoning)	81.6	91	57.4
StageGuard	88.4	87	67.1

Task SR = full long-horizon episode success rate on BEHAVIOR-1K simulation. StageGuard improves task success by 9.7 points over the best prior method (VLM FT) at essentially the same inference latency.

Ablations and Interpretation

Removing state layer: F1 drops 4.1 points; spatial relationship reasoning degrades without the structured state summary.

Removing task layer: F1 drops 3.6 points; subtask completion criteria are not grounded without explicit task-plan mapping.

Unstructured distillation: F1 = 83.9 (vs. 88.4); unstructured teacher traces dilute relevant decision factors with spurious reasoning.

History window H : $H = 5$ is optimal; $H = 1$ loses 4.8 F1; $H = 10$ gains 0.4 F1 at double the latency.

Student size: 3B achieves full StageGuard F1; 1B loses 3.2 F1; 7B adds only 0.5 F1 at $3\times$ latency.

Reference

Wenbo Cui, Jing Xu, Rui Zhao, Yunchao Liu, Chaobin Ye, Xinyu Zhang, Minghui Tan. **StageGuard: Learning Stage Transitions for Long-Horizon Robot Tasks via Agentic Distillation.** arXiv:2609.20791, September 2026. <https://arxiv.org/abs/2609.20791>

Keywords: agricultural AI • pixel-level grounding • multi-modal large language model • plant disease recognition

AgriScope

A Pixel-Grounded Multimodal Model and 11-Million-Sample Dataset Bring Precision to Agricultural Image Understanding

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Agricultural AI gets its first unified pixel-grounded multimodal model. *AgriScope* jointly supports image-level classification, region-level referring segmentation, and pixel-level grounded caption generation within a single framework trained on AgriGround—a new 500K-image, 11-million-sample dataset—lifting grounded caption CIDEr from 107.9 to 118.6 and mIoU from 55.0 to 59.4 over the best prior baseline on the accompanying benchmark.

AT A GLANCE

Problem: Agricultural image understanding requires fine-grained pixel-level localisation (disease lesions, pest bodies, growth stage features) that existing text-only MLLMs cannot provide.

Why it matters: Actionable recommendations for precision agriculture (targeted spraying, selective harvesting) require knowing *where* in the image the problem is, not just *what* it is.

Method: Biological-semantic encoder pre-trained on botanical taxonomies; multi-scale dense spatial features (FPN over BSE and DINOv2); mask transformer pixel decoder; AgriGround instruction- tuning dataset with 500K images and 11M samples.

Result: GCG CIDEr 118.6 (+10.7), GCG mIoU 59.4 (+4.4), referring seg F1 73.2 (+5.4) over PixelLM baseline.

Evidence: ICML 2026 evaluation across 6 agricultural task types; ablations over encoder, decoder, and dataset scale.

The Big Picture

Precision agriculture aims to deliver the right intervention (pesticide, fertiliser, pruning) to exactly the right location at exactly the right growth stage. This requires a visual understanding system that can answer not just “does this plant have rust disease?” but “which leaves show early-stage rust and where on each leaf are the uredinia?”

Multimodal large language models have advanced rapidly but their outputs are purely textual: they name diseases

and pests but do not localise them in pixel coordinates. AgriScope closes this gap with a unified model that takes an agricultural image (or image sequence) plus a natural-language instruction and outputs both a text response *and* pixel-level segmentation masks for all referenced visual entities.

Why Existing Approaches Fall Short

Text-only MLLMs (GPT-4V, InternVL-2, LLaVA) produce written diagnoses but cannot drive autonomous machinery that requires pixel masks or bounding boxes. Their agricultural knowledge is also limited by training on general web data rather than domain-curated imagery.

Specialist agricultural classifiers (PlantVillage CNNs, domain-specific ViTs) excel at narrow tasks (identify this disease from a leaf patch) but cannot answer open-ended agricultural questions, handle novel species, or produce grounded multi-entity outputs.

Grounding MLLMs (LISA, PixelLM) support referring segmentation for general-domain images but use standard ViT features that lack biological specificity: they cannot distinguish powdery mildew from leaf wax deposits, or aphid clusters from normal trichomes.

Core Mechanism

Biological-Semantic Encoder (BSE). A ViT-L fine-tuned on 2.3M agricultural images annotated with hierarchical biological labels (kingdom, class, order, family, genus, species, disease type, disease stage). The BSE produces feature tokens $\{f_i^{\text{bio}}\}$ embedding biological category memberships alongside spatial features.

Dense Spatial Representations (DSR). A multi-scale feature pyramid fuses BSE tokens with a frozen DINOv2 encoder:

$$F_{\text{DSR}} = \text{FPN}(f^{\text{bio}}, f^{\text{dino}}) \in \mathbb{R}^{C \times H/4 \times W/4},$$

providing spatially dense features at three resolutions for both coarse recognition and fine lesion localisation.

Pixel Decoder. A mask transformer decoder takes query embeddings $\{q_k\}$ derived from the LLM output’s [SEG] tokens and predicts binary masks:

$$M_k = \text{PD}(q_k, F_{\text{DSR}}) \in \{0, 1\}^{H \times W}.$$

When the LLM generates [SEG], the last hidden state at that position feeds directly into the pixel decoder as q_k .

Technical Formulation

Training minimises:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dice}} \mathcal{L}_{\text{dice}},$$

with $\lambda_{\text{BCE}} = 1.0$, $\lambda_{\text{dice}} = 0.5$.

Binary cross-entropy on predicted mask $\hat{M}_k$ vs. ground-truth M_k^* :

$$\mathcal{L}_{\text{BCE}} = - \sum_p [M_k^*(p) \log \hat{M}_k(p) + (1 - M_k^*(p)) \log (1 - \hat{M}_k(p))].$$

Dice loss for shape overlap:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_p M_k^*(p) \hat{M}_k(p)}{\sum_p M_k^*(p) + \sum_p \hat{M}_k(p)}.$$

Learning or Inference Procedure

1. Pre-train BSE on 2.3M agricultural images with hierarchical biological classification (warm-start from ViT-L ImageNet checkpoint).
2. Train AgriScope on AgriGround for 3 epochs with frozen DINOv2 and trainable BSE, LLM adapters (LoRA), and pixel decoder.
3. Optional task-specific fine-tuning for 1 epoch on narrow subsets (e.g., disease detection only).
4. Inference: input image + instruction; LLM generates text interleaved with [SEG] tokens; pixel decoder predicts a mask for each [SEG].

AgriGround construction: 500K source images from PlantVillage, iNaturalist-Agriculture, FGVC-Plant, and CropDiseaseDB; captions generated by GPT-4V with botanical system prompts; grounded with Grounding-DINO + SAM2; 11M instruction-following samples across 6 task types (disease, pest, species, growth, weed, nutritional deficiency).

What the Guarantee Says

No formal guarantees. The ICML reviewers cited the breadth of the AgriGround evaluation: 6 task types, 3 cross-dataset generalisation settings, and both in-distribution and zero-shot conditions, providing strong empirical evidence that improvements are not evaluation artefacts.

Experimental Findings

Main benchmark: AgriGround-Test (10K held-out samples).

Method	CIDEr	mIoU	Seg F1	VQA Acc
LLaVA-1.5	82.3	43.7	—	71.4
InternVL-2	97.1	49.8	—	78.9
LISA	101.4	52.3	64.1	74.2
PixelLM	107.9	55.0	67.8	76.5
AgriScope	118.6	59.4	73.2	82.7

CIDEr and mIoU for grounded caption generation (GCG). LLaVA and InternVL-2 do not support pixel-level output (Seg F1 not applicable). AgriScope is the first model to achieve over 59 mIoU on this benchmark.

Ablations and Interpretation

Standard ViT encoder (no BSE): CIDEr -6.8 ; the BSE’s biological specificity is critical for fine-grained disease staging.

Single-resolution features (no FPN): mIoU -3.1 ; multi-scale features are needed to localise both large leaf lesions and small pest bodies.

Without AgriGround (existing datasets only): CIDEr -14.3 ; breadth and scale of agricultural supervision are the primary driver.

Dataset scaling: 100K samples $\rightarrow$ 250K: $+6.2$ CIDEr; 250K $\rightarrow$ 500K: $+2.1$ CIDEr. Diminishing returns but still positive, suggesting continued benefit from data collection.

Decoder variant: Mask transformer vs. FPN upsampling: $+2.7$ mIoU; mask transformer’s cross-attention to text queries is essential for language-conditioned segmentation.

Reference

Abderrahmene Boudiaf, Mohamad Alanssari, Irfan Husain, Sajid Javed. **AgriScope: Pixel-Grounded Multimodal Understanding for Agricultural Images**. arXiv:2609.20325, September 2026. ICML 2026. <https://arxiv.org/abs/2609.20325>

Keywords: multi-modal image generation • evaluation metrics • condition alignment • chain-of-evaluation

UFO

Atomized Chain-of-Evaluation Achieves 15% Better Human-Correlation for Multi-Modal Image Generation by Evaluating All Conditions Simultaneously

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Multi-modal image generation is evaluated by checking each condition independently—a fundamentally wrong strategy when conditions interact and conflict. *UFO* introduces Atomized Chain-of-Evaluation: decompose each condition into fine-grained Atomic Evaluation Units, order them by dependency, call purpose-built verifiers, and aggregate. On the new UFO-Bench (660 cases, 7 categories, 3 difficulty tiers), *UFO* achieves Spearman $\rho=0.689$ with human rankings—a 15.25% improvement over the best prior metric.

AT A GLANCE

Problem: Existing multi-modal generation metrics evaluate each condition (text, reference image, layout, style, identity) in isolation, ignoring interactions that determine human preference.

Why it matters: Subject-driven customisation and layout- conditioned generation are competitive research areas; unreliable metrics stall progress and misrank models.

Method: Decompose each condition into Atomic Evaluation Units (AEUs); categorise by modality-relevance class; order into a chain; call structured (detector-based) or generic (MLLM) verifiers per AEU.

Result: Spearman $\rho = 0.689$ vs. $\rho = 0.598$ (VIEScore); +15.25% average improvement. Largest gain on hard (conflicting-condition) tier.

Evidence: UFO-Bench: 660 human-annotated cases, 5 raters each, Krippendorff $\alpha=0.73$; evaluated on CLIP-T, DINO-I, HPS-v2, VIEScore, and six other baselines.

The Big Picture

Subject-driven text-to-image models accept a reference image (the subject), a text prompt, a layout specification, and optionally a style or identity condition—all simultaneously. The output is evaluated for how well it aligns with all provided conditions at once.

Current metrics fail this task. CLIP-T measures text-image cosine similarity; DINO-I measures instance identity similarity; they are then averaged into a scalar. But this average hides condition priority: if a prompt says “make the dog hold a red umbrella” and the model produces a dog without an umbrella, no averaging of text and identity scores captures this failure correctly. Humans, presented with the same output, immediately rank it below a generation that at least includes the umbrella.

UFO treats evaluation as a structured reasoning problem: enumerate what must be verified, verify each part with the best available tool, and combine respecting priority and dependency.

Why Existing Approaches Fall Short

Embedding-based metrics (CLIP-T, DINO-I, CLIP-I) measure condition-specific cosine similarity in a shared embedding space. Their scores are invariant to fine-grained compositional failures; a tiger wearing a hat and a tiger without a hat may have nearly identical CLIP scores with the same text prompt.

MLLM-based metrics (VIEScore, Pick-Score) use a multimodal language model as a judge but ask it to evaluate all conditions holistically in a single prompt. The model must internally reason about interactions it was not specifically trained to handle; its judgments are inconsistent on conflicting-condition cases.

Human Preference Score (HPS-v2) is trained on general human preferences that reflect aesthetics and realism as much as condition alignment, making it poorly calibrated for the specific task of measuring omni-condition satisfaction.

Core Mechanism

AEU decomposition. For a set of conditions $C = \{c_1, \dots, c_K\}$ (text prompt c_T , reference image c_I , layout c_L , style c_S , ...), each condition is parsed into binary claims about one attribute:

$$\{u_{k,j}\} = \text{Parse}(c_k), \quad u_{k,j} : \text{“attribute } a_{k,j} \text{ of } c_k \text{ satisfied?”}$$

Chain ordering. AEUs are sorted topologically over a dependency graph G_C where edges encode precedence (subject identity must be located before subject pose can be verified). Mandatory conditions come first; preferred and contextual follow.

Specialised verifiers.

- *Text attributes:* MLLM with targeted yes/no prompts.
- *Identity conditions:* ArcFace (persons) or DINOv2 cosine similarity (objects).
- *Geometric conditions:* DWPose (body pose), Grounding-DINO + SAM2 (object placement),

DepthAnything (depth layout).

- *Style conditions*: CLIP style embedding vs. reference.

Weighted aggregation. Mandatory AEU’s have weight 1.0, preferred 0.7, contextual 0.3:

$$\text{OCA}_{\text{UFO}}(x, C) = \sum_{k,j} r_{k,j} \cdot V_{k,j}(x, c_k, a_{k,j}).$$

Technical Formulation

Let $\mathcal{U} = [u_1, \dots, u_n]$ be the ordered chain of AEU’s and V_i the verifier for AEU u_i . The OCA score is:

$$\text{OCA}(x, C) = \frac{\sum_{i=1}^n r_i V_i(x)}{\sum_{i=1}^n r_i}, \quad r_i \in \{1.0, 0.7, 0.3\}.$$

Spearman correlation with human rankings:

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)},$$

where d_i is the rank difference between UFO scores and human preference rankings over the $n = 660$ UFO-Bench cases.

For text prompt c_T with P phrases, each phrase p is parsed into an AEU with attribute type (object, property, spatial, action, count) and relevance class. The dependency graph adds edges from objects to their attributes (the object must be localised before its colour or pose can be verified).

Learning or Inference Procedure

UFO is training-free. Inference steps:

1. Parse all conditions C into AEU’s; extract attribute types and relevance classes.
2. Build dependency graph G_C ; topologically sort into chain $\mathcal{U}$.
3. For each AEU u_i : call verifier V_i with the generated image x and condition c_k .
4. Compute weighted aggregation $\text{OCA}_{\text{UFO}}(x, C)$.

New condition types are added by registering an AEU parser and verifier function—no retraining. Total evaluation time per image is approximately 4 seconds on a single GPU.

What the Guarantee Says

UFO provides a theoretical lemma: under conditional independence between conditions, the chain-of-evaluation score is an unbiased estimator of the true omni-condition alignment. For the general (dependent) case, the mandatory-first ordering minimises the expected rank correlation error by prioritising the conditions humans weight most heavily, which the paper verifies empirically across annotation tiers.

Experimental Findings

Rank correlation with human preferences on UFO-Bench (660 test cases, 5 annotators each):

Metric	Spearman ρ	Pearson r	Kendall τ
CLIP-T	0.421	0.398	0.312
DINO-I	0.389	0.374	0.288
HPS-v2	0.521	0.508	0.401
VIEScore	0.598	0.581	0.463
UFO	0.689	0.672	0.537

By difficulty tier: Easy $\rho = 0.741$; Medium $\rho = 0.693$; Hard $\rho = 0.701$. UFO’s advantage is largest on hard cases, where conflicting conditions are the norm and independent metrics collapse.

Ablations and Interpretation

MLLM for all AEU’s (no structured verifiers): ρ drops to 0.631; geometric and identity AEU’s are specifically hurt, confirming that dedicated detectors are needed for these types.

Random AEU ordering: $\rho = 0.657$; mandatory-first ordering resolves conflicting conditions more consistently.

Uniform weights ($r_i=1$ for all): $\rho = 0.672$; graded weights align with human priority on medium and hard tiers.

Sentence-level vs. phrase-level AEU’s: Phrase-level AEU’s improve ρ by 0.042; sub-phrase (word-level) adds only 0.003 at $2\times$ compute.

Model comparison on UFO-Bench: FLUX.1 outranks SDXL by 0.08 UFO score on hard cases but only 0.02 by CLIP-T average, illustrating UFO’s superior discriminative power for state-of-the-art models that are close to each other on simple metrics.

Reference

Danning Zhang, Yijing Lin, Shuhan Zhuang, Mengqi Huang, Shaojin Wu, Shancheng Fang, Zhendong Mao. **UFO: Chain-of-Evaluation for Omni-Condition Alignment in Multi-Modal Image Generation.** arXiv:2609.12397, September 2026. ICML 2026. <https://arxiv.org/abs/2609.12397>